Aditya Saini

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Education

Purdue University West Lafayette, IN
B.S. Computer Science (Systems Programming) *Graduation: May 2028 (Expected)*
Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Mathematics, Computer Architecture, Programming in C, Object-Oriented Programming, Multivariate Calculus

Experience

Embedded Engineer Sep. 2025 – Present
Purdue Electric Racing *West Lafayette, IN*

- Developed modular RTOS-based vehicle firmware in C, integrating drivers and performing in-circuit debugging
- Implemented runtime telemetry and data acquisition to validate performance bottlenecks in production firmware
- Designed I2C and SPI HALs for STM32 platforms, standardizing high-frequency sensor communication
- Developed Nextion LCD UART drivers and optimized command encoding for reliable real-time dashboard updates

Chip Design Member (SoCET STARS) June 2026 – August 2026
SoCET, Purdue VIP *West Lafayette, IN*

- Designed combinational and sequential logic (decoders, registers, synchronizers, FSMs) in SystemVerilog RTL and validated each block with self-checking testbenches
- Progressed through the full digital ASIC design flow from RTL to GDSII with OpenLane, applying FPGA synthesis, timing analysis, and MMIO/driver/accelerator integration
- Wrote RISC-V assembly for the AFTx07 ISA, simulated gate-level circuits in LTSPICE, and managed RTL builds with Linux, Makefiles, and Git on the asicfab server

Robotic Safety Research Assistant January 2026 – July 2026
PurSec, Purdue Computer Science *West Lafayette, IN*

- Researched jailbreak vulnerabilities in MLLM-controlled home-assistant robots, uncovering multi-step physics attacks that chain individually-safe actions into unsafe outcomes to defeat existing safety systems
- Engineered 10+ AI2-THOR multi-step physics attacks (chain-reaction, fire, and flooding hazards) that bypass state-of-the-art guardrails while every intermediate step passes inspection
- Reproduced a dual-brain safety baseline (RoboGuard: GPT-4o rules to Linear Temporal Logic in Spot Büchi automata; ASIMOV: GPT-4o Vision over frames + a Robot Constitution) and demonstrated attacks defeating both brains via a Dockerized headless pipeline on AWS EC2

Projects

Natural Language Shell | *C, Python, OpenAI API, Git* June 2025 – January 2026

- Created a lightweight command-line shell in C supporting natural language commands with tokenizer, intent parsing, and bash mapping; integrated OpenAI API for complex commands/queries through OpenRouter.

FRC Team 2554 | *Java, OpenCV, Jetson Orin Nano, CUDA* September 2021 – August 2025

- Led the team to its best-ever season as Build President, bringing us to District Championships after 18+ years.
- Developed and led our team's first instance of Swerve Drive, tuned PID/FF loops, and developed auto commands to increase accuracy by 66% using custom vision pipelines to detect AprilTags for robot positioning.

Tava | *React Native, Expo, TypeScript, Firebase, Mapbox, OpenAI* September 2025

- Built a cross-platform social app connecting student builders via an AI-powered search and Tinder-style swipe matching, with a node-edge network graph and interactive Mapbox map backed by Firebase auth and data.

Technical Skills

Languages: C, C++, SystemVerilog, Python, Java, SQL, JavaScript, HTML/CSS, R
Digital/Hardware: SystemVerilog RTL, Combinational & Sequential Logic, FSMs, Testbenching/Verification, RISC-V (AFTx07), FPGA Synthesis & Timing, RTL-to-GDSII (OpenLane), LTSPICE, ASIC Design Flow
Embedded: RTOS, STM32, ESP32, SPI, I2C, UART, CAN, TCP/IP, Arduino, Linux
Developer Tools: Git, Docker, Makefiles, AWS, Claude Code, Cursor, Agent Orchestration
Libraries: pandas, NumPy, Matplotlib, React, Node.js, PyTorch, CUDA, OpenCV